Representation Theory of Finite Groups

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March 10, 2026

Abstract

Contents

1 Representation of Groups

1.1 Basic Definitions

Definition 1.1.1 (Representations)
Let G be a group. Let k be a field. A representation of G is a k-vector space V together with a group homomorphism
$$\rho : G \to \text{End}_k(V)$$

Example 1.1.2
The following are representations of $D_8 = \langle r, s \mid r^4, s^2, rsrs \rangle$.
- $\rho : D_8 \to GL_2(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- $\rho : D_8 \to GL_n(\mathbb{C})$ defined by $\rho(r) = \rho(s) = I_n$.
- $\rho : D_8 \to \mathbb{C}^\times$ defined by $\rho(r) = \pm 1$ and $\rho(s) = \pm 1$.
- $\rho : D_8 \to GL_4(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

Lemma 1.1.3
Let G be a group. Let k be a field. Then there is a one to one correspondence
$$\left\{ \rho : G \to \text{End}_k(V) \;\middle|\; \substack{\rho \text{ is a} \\ \text{representation of } G} \right\} \overset{1:1}{\longleftrightarrow} \left\{ \rho : k[G] \to \text{End}_k(V) \;\middle|\; \substack{\rho \text{ is a } k\text{-algebra} \\ \text{homomorphism}} \right\}$$
given by the universal property of the group ring $k[G]$.

Lemma 1.1.4
Let G be a group. Let k be a field. Then there is a one to one correspondence
$$\left\{ \rho : k[G] \to \text{End}_k(V) \;\middle|\; \substack{\rho \text{ is a } k\text{-algebra} \\ \text{homomorphism}} \right\} \overset{1:1}{\longleftrightarrow} \left\{ f : k[G] \times V \to V \;\middle|\; \substack{f \text{ defines an } k[G]\text{-module} \\ \text{structure on } V} \right\}$$
given by $\rho \mapsto ((g, v) \mapsto \rho(g)(v))$.

Definition 1.1.5 (Types of Representations)
Let G be a group. Let k be a field. Let V be a representation of G.
- We say that V is irreducible if it is simple as a $k[G]$-module.
- We say that V is semisimple if it is semisimple as a $k[G]$-module.
- We say that V is indecomposable if it is indecomposable as a $k[G]$-module.

Example 1.1.6
The representation of $C_3 = \langle x \mid x^3 \rangle$ on $\mathbb{F}_3^2$ given by
$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$
is not semisimple.

Definition 1.1.7 (Composition Factors)
Let G be a group. Let k be a field. Let V be a semisimple representation of G. Define the composition factors of V to be
$$\text{Comp}(V) = \{V_1, \dots, V_n\}$$

where the irreducible decomposition of V is given by

$$V \cong \bigoplus_{i=1}^{n} V_i$$

Definition 1.1.8 (Multiplicity of an Irreducible Submodule)
Let G be a group. Let k be a field. Let V be a semisimple representation of G. Let W be an irreducible representation of V. Define the multiplicity of W in V to be

$$\text{mult}_W(V) = |\{V_i \in \text{Comp}(V) \mid V_i \cong W \text{ as } k[G]\text{-modules }\}|$$

Definition 1.1.9 (The Set of (Irreducible) Representations)
Let G be a group. Let k be a field.

- Define the set of representations of G by

$$\text{Rep}(G) = \{[V] \mid V \text{ is a representation of } G\}$$

- Define the set of representations of G by

$$\text{IrrRep}(G) = \{[V] \mid V \text{ is an irreducible representation of } G\}$$

Definition 1.1.10 (The Regular Representations)
Let G be a group. Let k be a field. Define the regular representation of G to be the function $\text{reg}_G : G \to \text{End}_k(k[G])$ given by

$$\rho(g)(h) = g \cdot h$$

Thus the regular representation of G is just the $k[G]$-algebra $k[G]$.

Definition 1.1.11 (The Permutation Representations)
Let X be a set. Let G be a group acting on X. Let k be a field. Define the permutation representation $\rho : G \to \text{GL}(k\langle X\rangle)$ of G on X over k by the formula

$$\rho(g)(x) = g \cdot x$$

and then extending it k-linearly.

The regular representation is just the permutation representation with the group action on itself.

Definition 1.1.12 (G-Invariant Subspaces)
Let G be a group. Let k be a field. Let V be a representation of G. Define the G-invariant subspace of V to be

$$V^G = \{v \in V \mid \rho(g)(v) = v \text{ for all } g \in G\}$$

If we think of representations as actions of $k[G]$ on V and hence actions of G on V, then V^G is just the fixed points of the action.

1.2 Maps between Representations

Definition 1.2.1 (Homomorphisms of Representations)
Let G be a group. Let k be a field. Let $\rho : G \to \text{End}_k(V)$ and $\theta : G \to \text{End}_k(W)$ be representations of G. Let $f : V \to W$ be a function. We say that f is a homomorphism of representations if the following are true.

- f is a k-linear map.
- $f \circ \rho(g) = \theta(g) \circ f$ for all $g \in G$.

Lemma 1.2.2
Let G be a group. Let k be a field. Let V, W be representations of G. Let $f : V \to W$ be a function. Then f is a homomorphism of representations if and only if f is a $k[G]$-algebra homomorphism.

Definition 1.2.3 (Isomorphisms of Representations)
Let G be a group. Let k be a field. Let V, W be representations of G. Let $f : V \to W$ be a homomorphism. We say that f is an isomorphism if there exists a homomorphism $g : W \to V$ such that

$$g \circ f = \mathrm{id}_V \quad \text{and} \quad f \circ g = \mathrm{id}_W$$

In this case we say that V and W are isomorphic representations.

Lemma 1.2.4
Let G be a group. Let k be a field. Let V, W be representations of G. Let $f : V \to W$ be a homomorphism. Then f is an isomorphism if and only if f is an isomorphism of k-vector spaces.

Let R be a ring. Let M, N be R-modules. Recall that in general there is no canonical module structure for $\mathrm{Hom}_R(M, N)$ unless R is commutative. Hence $\mathrm{Hom}_{k[G]}(V, W)$ is a priori not a representation of G. However, we will give a module structure to it by giving a $k[G]$-module structure to $\mathrm{Hom}_k(V, W)$ and identifying $\mathrm{Hom}_{k[G]}(V, W)$ with $\mathrm{Hom}_k(V, W)^G$.

Lemma 1.2.5
Let G be a group. Let k be a field. Let $\rho_V : G \to \mathrm{End}_k(V)$ and $\rho_W : G \to \mathrm{End}_k(W)$ be representations of G. Then $\mathrm{Hom}_k(V, W)$ is also a representation of G via the map $g \mapsto \rho_W(g) \circ - \circ \rho_V(g)^{-1}$.

Proposition 1.2.6
Let G be a group. Let k be a field. Let V, W be representations of G. Then there is a vector space isomorphism

$$\mathrm{Hom}_{k[G]}(V, W) \cong \mathrm{Hom}_k(V, W)^G$$

given by restriction of scalars.

Proof
Let $f : V \to W$ be a $k[G]$-module homomorphism. We claim that after restricting to k, we have $f \in \mathrm{Hom}_k(V, W)^G$. Indeed, since f is a homomorphism of representations, we have $\rho_W(g) \circ f = f \circ \rho_V(g)$ for all $g \in G$. This implies that $\rho_{\mathrm{Hom}_k(V,W)}(g)(f) = \rho_W(g) \circ f \circ \rho_V(g)^{-1} = f$. Hence $f \in \mathrm{Hom}_k(V, W)^G$. Hence we have a well defined k-linear map $\mathrm{Hom}_{k[G]}(V, W) \to \mathrm{Hom}_k(V, W)^G$. This map is clearly injective and so is a vector space isomorphism. ■

Note that the above isomorphism of vector spaces give $\mathrm{Hom}_{k[G]}(V, W)$ the structure of a $k[G]$-module.

1.3 One Dimensional Representations

Let G be a group. A one dimensional representation of G is a group homomorphism $G \to k^\times$. Since $k^\times$ is abelian, we may invoke the universal property of the abelianization G^{ab} in Advanced Group Theory to see that there is a one to one bijection between one dimensional representations of G and one dimensional representations G^{ab}.

Lemma 1.3.1
Let G be a group. If G^{ab} is finite, then the number of one dimensional representations of G is G^{ab}.

Let G be a group. Recall that G is perfect if $G = [G, G]$. Equivalently, this is the same as saying that $G^{\text{ab}} = 0$.

Lemma 1.3.2
Let G be a group. If G is perfect, then there are no non-trivial one dimensional representations of G.

Thus simple groups have no non-trivial one dimensional representations.

Proposition 1.3.3
Let G be a group. Let k be a field. Let $\rho : G \to \text{End}_k(V)$ be a representation of G. Let $\theta : G \to k^\times$ be a one dimensional representation of G. Then the following are true.
- $\theta\rho : G \to \text{End}_k(V)$ is also a representation of G.
- $\theta\rho$ is irreducible if and only if ρ is irreducible.
- $\theta\rho$ is the tensor product of θ and ρ as $k[G]$-modules.

Proof
Let $g, h \in G$. Then we have

$$(\theta\rho)(gh) = \theta(gh)\rho(gh) = \theta(g)\theta(h)\rho(g)\rho(h) = \theta(g)\rho(g)\theta(h)\theta(h) = (\theta\rho)(g)(\theta\rho)(h)$$

since $\theta(h) \in k^\times$ is a scalar. Hence $\theta\rho$ is a representation.

Suppose that ρ has a subrepresentation W. Then for all $w \in W$, we have $(\theta\rho)(g)(w) = \theta(g)(w)\rho(g)(w) \in W$ since $\theta(g)(w) \in k^\times$ is a scalar and W is a vector space. Hence W is a subrepresentation of $\theta\rho$. Conversely, notice that if $\theta : G \to k^\times$ is a representation, then $(-)^{-1} \circ \theta : G \to k^\times$ is also a one dimensional representation. Thus if W is a subrepresentation of $\theta\rho$, then W is a subrepresentation of $((-)^{-1} \circ \theta)\theta\rho = \rho$.

The third item follows from the fact that $k \otimes_k V \cong V$ as vector spaces. ■

1.4 Representations over Algebraically Closed Fields

Lemma 1.4.1 (Schur's Lemma III)
Let G be a group. Let k be an algebraically closed field. Let V, W be irreducible representations of G. Let $\phi : V \to W$ be a homomorphism. Then the following are true.
- If $V \cong W$ and they are finite dimensional, then $\phi = \lambda \text{id}_V$ for some $\lambda \in k$.
- If $V \ncong W$, then $\phi = 0$.

Corollary 1.4.2
Let G be a group. Let k be an algebraically closed field. Let V be a finite dimensional representations of G. If G is abelian, then $\dim_k(V) = 1$.

Corollary 1.4.3
Let G be a group. Let k be an algebraically closed field. Let V, W be finite dimensional irreducible representations of G. Then we have

$$\dim_k(\text{Hom}_{k[G]}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Corollary 1.4.4
Let G be a group. Let k be an algebraically closed field. Let V, W be finite dimensional representations of G such that W is irreducible. Then we have

$$\dim_k(\text{Hom}_{k[G]}(V, W)) = \text{mult}_W(V)$$

1.5 Representations of Finite Groups over Algebraically Closed Fields

Landmark result: Maschke's theorem. Recall that it says that $k[G]$ is a semisimple k-algebra if G is finite and $\text{char}(k)$ does not divide $|G|$. By a result in Rings and Modules, this implies that every $k[G]$-module is semisimple. Hence every representation of G is semisimple.

Proposition 1.5.1
Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Suppose that

$$k[G] \cong \bigoplus_{i=1}^{n} \text{End}_k(V_i)$$

be the decomposition given by the Artin-Wedderburn theorem. Then we have

$$\text{IrrRep}(G) = \{[V_1], \dots, [V_n]\}$$

Proof
Follows from a corollary of Artin-Wedderburn theorem in Advanced Ring Theory. ■

Corollary 1.5.2
Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Then we have

$$|G| = \sum_{V \in \text{IrrRep}(G)} (\dim_k(V))^2$$

Proof
Follows from the above proposition. ■

Example 1.5.3
Let G be a group of order 8. Let k be an algebraically closed field such that $\text{char}(k) \neq 2$. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $k[G]$-modules. Then exactly one of the following is true (up to reordering of the modules).

- $n = 8$ and $\dim(V_i) = 1$.
- $n = 5$ and $\dim(V_1) = \cdots = \dim(V_4) = 1$ and $\dim(V_5) = 2$.

Corollary 1.5.4
Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. If $[V] \in \text{IrrRep}(G)$ is an irreducible representation, then we have

$$\dim_k(V) < \sqrt{|G|}$$

Proof
We have

$$|G| = \sum_{V \in \text{IrrRep}(G)} (\dim_k(V))^2 > \dim_k(V)^2$$

where the inequality is strict because the trivial representation is irreducible and always exists. ■

2 Character Theory

2.1 Basic Definitions

Definition 2.1.1 (Characters)
Let G be a finite group. Let k be an algebraically closed field. Let $\rho : G \to \text{End}_k(V)$ be a finite dimensional representation of G. Define the character of V to be the function $\chi_V : G \to k$ given by

$$\chi_V(g) = \text{tr}(\rho(g))$$

Proposition 2.1.2
Let G be a finite group. Let k be an algebraically closed field such that char(k) does not divide $|G|$. Let $\rho : G \to \text{End}_k(V)$ be a representation of G. Then the following are true.

- $\chi_V(1_G) = \dim(V)$.
- $\chi_V(hgh^{-1}) = \chi(g)$ for all $g, h \in G$.

Proof
By Linear Algebra, we have that

$$\chi_V(1_G) = \text{tr}(\rho(1_G)) = \text{tr}(\text{id}_V) = \text{tr}(I_{\dim(V)}) = \dim(V)$$

so $\chi_V(1_G) = \dim(V)$.

Let $g, h \in G$. Then we have

$$\chi_V(hgh^{-1}) = \text{tr}(\rho(h)\rho(g)\rho(h^{-1})) = \text{tr}(\rho(h)\rho(h^{-1})\rho(g)) = \text{tr}(\rho(hh^{-1})\rho(g)) = \text{tr}(\rho(g)) = \chi_V(g)$$

by properties of the trace map. ∎

The second item implies that χ descends to a function on conjugation classes $\chi : \text{Cl}(G) \to k$.

Lemma 2.1.3
Let G be a finite group. Let k be an algebraically closed field. Then χ descends to a well defined class function $\chi : \text{Cl}(G) \to k$.

Lemma 2.1.4
Let G be a finite group. Let k be an algebraically closed field. Let V, W be finite dimensional representations of G. Then the following are true.

- If $W \leq V$ is a subrepresentation, then $\chi_V = \chi_W + \chi_{V/W}$.
- $\chi_{V \otimes W} = \chi_V \chi_W$.
- $\chi_{V^*} = \overline{\chi}$.
- $\chi_{\text{Hom}_k(V,W)} = \overline{\chi_V}\chi_W$.

Lemma 2.1.5
Let G be a finite group. Let k be an algebraically closed field. Let V, W be finite dimensional representations of G. If $V \cong W$ are isomorphic, then we have

$$\chi_V = \chi_W$$

Proposition 2.1.6
Let G be a finite group. Let $\rho : G \to \text{End}_{\mathbb{C}}(V)$ be a complex representation of G. Let $g \in G$. Then the following are true.

- Let ω be a $|g|$th root of unit. Then we have

$$\chi(g) = \sum_{i=1}^{k} \omega^{j_i}$$

for some $j_i \in \mathbb{N}$ and $k \in \mathbb{N}$.

- We have $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Proof
Let $H = \langle g \rangle$. Then V is also a $\mathbb{C}[H]$-module. Let $V = \bigoplus_{i=1}^m V_i$ be the decomposition of V into irreducible $\mathbb{C}[H]$-submodules. Since H is cyclic, we must have $\dim(V_i) = 1$ by crl2.1.2. Choose non-zero $v_i \in V_i$. Then $E = \{v_1, \dots, v_n\}$ is a basis for V and $[\rho(g)]_E^E$ is a diagonal matrix. Now for each i, V_i is a $\mathbb{C}[H]$-module. Hence $\rho(g)(v_i) \in V_i$ implies that $\rho(g)(v_i) = \lambda_i v_i$ for some $\lambda_i \in \mathbb{C}^\times$. Then we have $\lambda^{|g|} v_i = \rho(g)^{|g|}(v_i) = v_i$ so that λ is a $|g|$th root of unity. Since $\rho(g)$ is diagonal, we conclude that the trace of $\rho(g)$ is a sum of $|g|$th roots of unity.

From the above proof, we know that $[\rho(g)]_E^E = \text{diag}(\omega^{-j_1}, \dots, \omega^{-j_m}) = \text{diag}(\overline{\omega^{j_1}}, \dots, \overline{\omega^{j_m}})$. Hence $\chi_V(g^{-1}) = \overline{\chi_V(g)}$. ■

Definition 2.1.7 (Irreducible Characters)
Let G be a finite group. Let k be an algebraically closed field. Let V be a finite dimensional representation of G. We say that χ_V is irreducible if V is irreducible.

Definition 2.1.8 (Characters of a Group)
Let G be a finite group. Let k be an algebraically closed field. Let $\chi : G \to k$ be a function. We say that χ is a character of G if there exists a finite dimensional representation V of G such that $\chi = \chi_V$.

2.2 Orthogonality of Characters of Irreducible Representations

Definition 2.2.1 (The Inner Product Space of Class Functions)
Let G be a finite group. Define the inner product space of class functions of G to be the $\mathbb{C}$-vector space

$$\text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})$$

together with the inner product given by

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

for $f_1, f_2 \in \text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})$.

Proposition 2.2.2
Let G be a finite group. Let V, W be finite representations of G over $\mathbb{C}$. Then we have

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}[G]}(V, W))$$

Proof
Step 1:
Let $T : V \to V^G$ be the map defined by $v \mapsto \frac{1}{|G|} \sum_{g \in G} \rho(g)(v)$. I claim that $\text{tr}(T) = \dim_{\mathbb{C}}(V^G)$. Notice that since V^G is G-invariant, we have that $T^2 = T$. I claim that $V^G = \ker(T) \oplus \text{im}(T)$. Suppose that $v \in \ker(T) \cap \text{im}(T)$. Then there exists $w \in V$ such that $T(w) = v$. Now $v = T(w) = T^2(w) = T(T(w)) = T(v) = 0$. Hence $\ker(T)$ and $\text{im}(T)$ has empty intersection. Now let $v \in V$. Then we have $v = (v - T(v)) + T(v)$. The latter lies in $\text{im}(T)$ and the former lies in $\ker(T)$ since $T(v - T(v)) = T(v) - T(T(v)) = T(v) - T(v) = 0$. Hence $V = \ker(T) \oplus \text{im}(T)$. Choose a basis $b_1, \dots, b_k$ for $\text{im}(T)$, choose a basis for $b_{k+1}, \dots, b_n$ for $\ker(T)$. Since $V = \ker(T) \oplus \text{im}(T)$, matrix for T under the basis $b_1, \dots, b_n$ is given by $\begin{pmatrix} I_{\dim(\text{im}(T))} & 0 \\ 0 & 0 \end{pmatrix}$. Hence $\text{tr}(T) = \dim(\text{im}(T)) = \dim(V^G)$.

Step 2:

We have

$$\langle \chi_{\text{triv}}, \chi_V \rangle = \frac{1}{|G|}\sum_{g\in G}\overline{\chi_V(g)} = \frac{1}{|G|}\sum_{g\in G}\overline{\text{tr}(\rho(g))} = \text{tr}\left(\frac{1}{|G|}\sum_{g\in G}\overline{\rho(g)}\right) = \text{tr}(\overline{T}) = \overline{\text{tr}(T)} = \dim_{\mathbb{C}}(V^G)$$

Step 3:
We have

$$\langle \chi_V, \chi_W \rangle = \frac{1}{|G|}\sum_{g\in G}\chi_V(g)\overline{\chi_W(g)} = \frac{1}{|G|}\sum_{g\in G} 1\cdot\overline{\overline{\chi_V(g)}\chi_W(g)} = \langle 1, \overline{\chi_V}\chi_W\rangle = \langle \chi_{\text{triv}}, \chi_{\text{Hom}_{\mathbb{C}}(V,W)}\rangle$$

Step 4:
Combining all the above, we have

$$\langle \chi_V, \chi_W \rangle = \langle \chi_{\text{triv}}, \chi_{\text{Hom}_{\mathbb{C}}(V,W)}\rangle = \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}}(V,W)^G) = \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}[G]}(V,W))$$

Corollary 2.2.3
Let G be a finite group. Let V, W be finite representations of G over $\mathbb{C}$. Then the following are true.

- If V and W are irreducible, then we have
$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$
- V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

Proof
Follows directly from the properties of $\dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}[G]}(V,W))$.

Proposition 2.2.4
Let G be a finite group. Then the following are true.

- We have $|\text{IrrRep}(G)| = \dim_{\mathbb{C}}(\text{Hom}_{\textbf{Set}}(\text{Cl}(G),\mathbb{C})) = |\text{Cl}(G)|$.
- The set $\{\chi_V \mid V \in \text{IrrRep}(G)\}$ is an orthonormal basis for $\text{Hom}_{\textbf{Set}}(\text{Cl}(G),\mathbb{C})$.

Proof
I claim that $\dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = |\text{Cl}(G)|$. Let $x = \sum_{h\in G}\lambda_h h \in \mathbb{C}[G]$. Notice that $xy = yx$ for all $y \in \mathbb{C}[G]$ if and only if $xg = gx$ for all $g \in G$ by linearity. Now notice that we have $gxg^{-1} = x$ if and only if

$$\sum_{h\in G}\lambda_h ghg^{-1} = \sum_{h\in G}\lambda_h h$$

By comparing basis for $\mathbb{C}[G]$, we see that this is true if and only if $\lambda_{g^{-1}hg} = \lambda_h$ for all $h \in G$. Thus $x = \sum_{h\in G}\lambda_h h \in Z(\mathbb{C}[G])$ if and only if $\lambda_{g^{-1}hg} = \lambda_h$ for all $h \in G$. This means that x is constant on conjugacy classes. Such a vector space has dimension $|\text{Cl}(G)|$.

Now by Artin-Wedderburn theorem, we have

$$\mathbb{C}[G] \cong \bigoplus_{V\in\text{IrrRep(G)}} \text{End}_{\mathbb{C}}(V)$$

Since taking center respects products, we have

$$Z(\mathbb{C}[G]) \cong \bigoplus_{V\in\text{IrrRep}(G)} Z(\text{End}_{\mathbb{C}}(V)) \cong \mathbb{C}^{|\text{IrrRep}(G)|}$$

since $\text{End}_{\mathbb{C}}(V) \cong M_{\dim(V)}(\mathbb{C})$ and from advanced ring theory we have seen that $Z(M_{\dim(V)}(\mathbb{C})) \cong \mathbb{C}$. Taking dimensions on both sides, we deduce that $|\text{Cl}(G)| = \dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = |\text{IrrRep}(G)|$.

Now since $\{\chi_V \mid V \in \text{IrrRep}(G)\}$ are orthogonal, they are linearly independent and has the same cardinality the dimension of $\text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})$. Hence they form an orthonormal basis. ∎

Corollary 2.2.5
Let G be a finite group. Let $g, h \in G$. Then g and h are conjugate if and only if $\chi(g) = \chi(h)$ for all characters χ of G.

Corollary 2.2.6
Let G be a finite group. Let $g \in G$. Then g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ of G.

2.3 Character Tables

Definition 2.3.1 (The Character Table of a Finite Group)
Let G be a finite group. Let $V_1, \dots, V_{|\text{Cl}(G)|}$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$-modules. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G. Define the character table of G to be the matrix $(a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ defined by

$$a_{i,j} = \chi_{V_i}(g_j)$$

Proposition 2.3.2
Let G be a finite group. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G. Let $A = (a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ be the character table of G. Then the following are true.

- The rows of A are orthogonal:

$$\frac{1}{|G|}\sum_{j=1}^{|\text{Cl}(G)|} |\text{Cl}(g_j)|\chi_{V_r}(g_j)\overline{\chi_{V_s}(g_j)} = \sum_{j=1}^{|\text{Cl}(G)|} |\text{Cl}(g_j)| a_{r,j}\overline{a_{s,j}} = \begin{cases} 1 & \text{if } r = s \\ 0 & \text{otherwise} \end{cases}$$

- The columns of A are orthogonal. This means that

$$\sum_{i=1}^{|\text{Cl}(G)|} \chi_{V_i}(g_r)\overline{\chi_{V_i}(g_s)} = \sum_{i=1}^{|\text{Cl}(G)|} a_{i,r}\overline{a_{i,s}} = \begin{cases} |\text{Cl}(g_r)| & \text{if } r = s \\ 0 & \text{otherwise} \end{cases}$$

A number of previous theorems help determining the character table. To compute the number of irreducible representations, we may use the following facts.

- $|G| = \sum_{V \in \text{IrrRep}(G)} \dim(V)^2$.
- $|\text{Cl}(G)| = |\text{IrrRep}(G)|$.

To then figure out the character table, we may use the following facts.

- $\chi_V(1_G) = \dim(V)$.
- The trivial map $G \to \mathbb{C}$ given by $g \mapsto 1$ is always a representation of G corresponding to the trivial representation.
- Orthogonality relations between the rows and the columns
- $\chi(g^{-1}) = \overline{\chi(g)}$ for all characters χ.
- g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ.
- If χ is an irreducible character, then $\overline{\chi}$ is also an irreducible character corresponding to the dual representation.
- If V is one-dimensional, then $\chi_V = \rho$ and χ is a group homomorphism.

Example 2.3.3
The character table for $D_6 = \langle r, s \mid r^3, s^2, rsrs \rangle$ is given by

D_6	1	$\{r, r^2\}$	$\{s, sr, sr^2\}$
χ_{Trivial}	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

where χ_2 is the character associated to the representation $D_6 \to \mathbb{C}$ given by $r \mapsto 1$, $s \mapsto -1$ and χ_3 is the character associated to the representation

Example 2.3.4
The character table for $D_8 = \langle r, s \mid r^4, s^2, rsrs \rangle$ is given by

D_8	1	$\{r, r^3\}$	$\{r^2\}$	$\{s, sr^2\}$	$\{sr, sr^3\}$
χ_{Trivial}	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	-1	1	-1	1
χ_5	2	0	-2	0	0

Example 2.3.5
The character table for A_4 is given by

A_4	1	$(1\ 2)(3\ 4)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
χ_{Trivial}	1	1	1	1
χ_2	1	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
χ_3	1	1	$e^{4\pi i/3}$	$e^{2\pi i/3}$
χ_3	3	-1	0	0

where χ_2 is the character associated to the representation $D_6 \to \mathbb{C}$ given by $r \mapsto 1$, $s \mapsto -1$ and χ_3 is the character associated to the representation

Example 2.3.6
The character table for C_n is given by the matrix:

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & \zeta & \cdots & \zeta^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \zeta^{n-1} & \cdots & (\zeta^{n-1})^{n-1} \end{pmatrix}$$

where $\zeta = e^{2\pi i/n}$.

2.4 Representation Theory of Subgroups

Recall that we have the notion of restriction of scalars and extension of scalars in Rings and Modules. Let G be a group. Let k be a field. Let $H \leq G$ be a subgroup.

- Let V be a representation of G. The restriction of V to H is be the k-vector space
$$\text{Res}_H^G(V) = V$$
together with the action of $k[H]$ given by restricting the action of $k[G]$ to $k[H]$.
- Let U be a representation of H. The extension of U to G is the k-vector space
$$\text{Ind}_H^G(U) = k[G] \otimes_{k[H]} U$$
together with the action of $k[G]$ given by the action on the component $k[G]$ in the tensor product.

Note: Even if V is an irreducible $k[G]$-module, it may not be an irreducible $k[H]$-module, and the reverse is also true.

Proposition 2.4.1
Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a non-zero character of H. Then there exists an irreducible character χ of G such that $\langle \text{Res}_H^G \chi, \psi \rangle \neq 0$.

Proof
Let $\chi_1, \dots, \chi_k$ be the complete list of irreducible characters of G. Let χ_{reg} be the character of the regular representation of G. Then we know that $\chi_{\text{reg}} = \sum_{i=1}^k \chi_i(1_G)\chi_i$. Notice that the regular $\mathbb{C}[G]$-module contains $\mathbb{C}[H]$ and $\mathbb{C}[H]$ is closed under the action of H. Hence $\mathbb{C}[H]$ is a submodule of $\text{Res}_H^G(\mathbb{C}[G])$.

Suppose that ψ is irreducible. Then the corresponding $\mathbb{C}[H]$-module V is a submodule of the regular representation $\mathbb{C}[H]$. Thus $\langle \text{Res}_H^G \chi_{\text{reg}}, \psi \rangle \neq 0$. Then we have

$$0 \neq \left\langle \sum_{i=1}^k \chi_i(1_G) \text{Res}_H^G \chi_i, \psi \right\rangle = \sum_{i=1}^k \chi_i(1_G) \langle \text{Res}_H^G \chi_i, \psi \rangle$$

Thus there exists χ_i such that $\langle \text{Res}_H^G \chi_i, \psi \rangle \neq 0$.

We can extend this to the case where ψ is reducible by noting that for a character ψ_j for some irreducible composition factor of ψ, we have $\langle \text{Res}_H^G \chi_i, \psi_j \rangle \neq 0$. ■

Theorem 2.4.2 (Clifford's Theorem)
Let G be a finite group. Let $H \trianglelefteq G$ be a normal subgroup. Let χ be an irreducible character of G. Let $\psi_1, \dots, \psi_n$ be the complete list of irreducible characters of H such that $\text{Res}_H^G \chi$ is a linear combination of $\psi_1, \dots, \psi_n$ with non-zero coefficients. Then the following are true.

- Each ψ_i have the same dimension.
- $\langle \text{Res}_H^G \chi, \psi_i \rangle$ is the same for all i.

Proof
Let V be an irreducible $\mathbb{C}[G]$-module corresponding to χ. Let U be an irreducible $\mathbb{C}[H]$-module of $\text{Res}_H^G(V)$. For all $g \in G$, the set $g \cdot U$ is a subspace of V. For all $h \in H$, we have

$$h(gu) = g(g^{-1}hg)u \in gU$$

since U is a $\mathbb{C}[H]$-module and H is normal implies $g^{-1}hg \in H$. So $g \cdot U$ is a $\mathbb{C}[H]$-submodule of $\text{Res}_H^G(V)$.

I claim that $g \cdot U$ is irreducible. Suppose that W is a $\mathbb{C}[H]$-submodule of $g \cdot U$. Then $g^{-1} \cdot W$ is a $\mathbb{C}[H]$-submodule of U by the above paragraph. Hence $g^{-1} \cdot W$ is U or 0 since U is irreducible. Hence W is equal to $g \cdot U$ or 0. Thus $g \cdot U$ is irreducible. Since $g \cdot -$ is an invertible linear map, we have $\dim_{\mathbb{C}}(g \cdot U) = \dim_{\mathbb{C}}(U)$.

Let $g_1, \dots, g_l$ be complete list of distinct elements in G such that $g_1 \cdot U, \dots, g_l \cdot U$ are pairwise not equal to each other. Then $\bigoplus_{j=1}^l g_j \cdot U$ is a $\mathbb{C}[G]$-submodule of V. Since V is an irreducible $\mathbb{C}[G]$-module, we have $\bigoplus_{g \in G} g \cdot U = V$. Hence V is the direct sum of these irreducible submodules of equal dimension. This proves part 1 of the theorem. ■

Proposition 2.4.3
Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be an irreducible character of G. Let $\psi_1, \dots, \psi_r$ be the complete list of irreducible characters of H. Suppose that $\text{Res}_H^G \chi = \sum_{i=1}^r d_i \psi_i$. Then the following are true.

- We have $\sum_{i=1}^r d_i^2 \leq [G : H]$.
- The above equality holds if and only if $\chi(g) = 0$ for all $g \in G \setminus H$.

Proof
Notice that
$$\sum_{i=1}^r d_i^2 = \langle \text{Res}_H^G \chi, \text{Res}_H^G \chi \rangle_H = \frac{1}{|H|} \sum_{h \in H} \chi(h)\overline{\chi(h)}$$
and also
$$1 = \langle \chi, \chi \rangle_G = \frac{1}{|G|} \left(\sum_{h \in H} \chi(h)\overline{\chi(h)} + \sum_{g \in G \setminus H} \chi(g)\overline{\chi(g)} \right) = \frac{|H|}{|G|} \left(\sum_{i=1}^r d_i^2 \right) + \frac{1}{|G|} \sum_{g \in G \setminus H} \chi(g)\overline{\chi(g)}$$
so we are done. ■

Lemma 2.4.4
Let G be a group. Let k be a field. Let $H \leq G$ be a subgroup. Let $g_1 H, \dots, g_n H$ be the complete list of distinct left cosets of H in G. Let U be a representation of H. Then we have
$$\text{Ind}_H^G(U) \cong \bigoplus_{i=1}^n g_i H$$

Example 2.4.5
Let $G = D_3 = \langle a, b \mid a^3, b^2, abab \rangle = S_3$. Let $H = \langle a \rangle \leq D_3$. Then the complete list of irreducible $\mathbb{C}[H]$-modules are given by

- $W_1 = \mathbb{C}\langle 1_H + a + a^2 \rangle$.
- $W_2 = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2 \rangle$.
- $W_3 = \mathbb{C}\langle 1_H + \omega^2 a + \omega a^2 \rangle$.

where $\omega = e^{2\pi i/3}$. Then the induced representations of the modules are given by

- $\text{Ind}_H^G(W_1) = \mathbb{C}\langle 1_H + a + a^2, b + ba + ba^2 \rangle$.
- $\text{Ind}_H^G(W_2) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.
- $\text{Ind}_H^G(W_3) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.

Recall that we have a change of rings adjunction formula:
$$\text{Hom}_{k[G]}(\text{Ind}_H^G(U), V) \cong \text{Hom}_{k[H]}(U, \text{Res}_H^G(V))$$

Theorem 2.4.6 (Frobenius Reciprocity Theorem)
Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be a character of G. Let ψ be a character of H. Then the change of base rings adjuncntion induces the following:
$$\langle \text{Ind}_H^G \psi, \chi \rangle_G = \langle \psi, \text{Res}_H^G \chi \rangle_H$$

Proposition 2.4.7 (The Mackey Formula)
Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a character of H. Then we have
$$\text{Ind}_H^G \psi(g) = \frac{1}{|H|} \sum_{y \in G, ygy^{-1} \in H} \psi(ygy^{-1})$$

3 Representation Theory of the Symmetric Group

3.1

The conjugacy classes of S_n is the given by the cycle type of the permutations. These are indexed by partitions $\lambda = (\lambda_1, \dots, \lambda_k)$ such that $\sum_{i=1}^k \lambda_i = n$ and $\lambda_i \geq \lambda_{i+1}$. We write $\lambda \vdash n$. These can be drawn into Young diagrams:

- There are λ_i boxes in row i.
- Rows are left justified.

Let S_λ be the subgroup of S_n given by

$$S_\lambda = S_{1,\dots,\lambda_1} \times S_{\lambda_1+1,\dots,\lambda_1+\lambda_2} \times \dots \times S_{n-\lambda_k+1,\dots,n} \cong S_{\lambda_1} \times \dots \times S_{\lambda_k}$$

Let M^λ be the module associated to the character $\text{Triv}_{S_\lambda}^{S_n}$. This is not generally irreducible. But we can order partitions of n such that the following are true.

- $M^{\lambda(1)}$ is irreducible (call it $S^{\lambda(1)}$)
- $M^{\lambda(2)}$ contain copies of $S^{\lambda(1)}$ and a unique copy of a new irreducible module $S^{\lambda(2)}$.
- etc...

$S^{\lambda(i)}$ are called Specht modules. $M^{\lambda(i)}$ are called the permutation modules corresponding to λ. The ordering needed is reverse-lexicographic (revlex). Largest number to smallest number, starting with the first component and moving to latter components to break ties.

Let $\overline{t_i}$ denote the distinct λ-tabloids: fill the young diagram of λ with $\{1, \dots, n\}$ (this is called a Young tableau) and mark two as equivalent if each corresponding row has the same elements. Then $M^\lambda = \mathbb{C}\langle \overline{t_1}, \dots, \overline{t_k} \rangle$.

Example 3.1.1
The partitions of $\{1, \dots, 4\}$ is given by

$$\{(4), (3,1), (2,2), (2,1,1), (1,1,1,1)\}$$

Example 3.1.2
Let $\lambda = (n)$. Then $M^\lambda \cong \mathbb{C}$ and S_n acts on this trivially. Thus $\text{Ind}_{S_\lambda}^{S_n} \text{Triv} = \text{Triv}$.

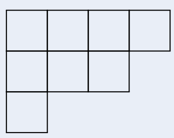
Let $\lambda = (1, \dots, 1)$. Then $M^\lambda \cong \mathbb{C}^{n!}$. Then $\text{Ind}_{S_\lambda}^{S_n} \text{Triv} = \chi_{\text{reg}}$.

Let $\lambda = (n-1, 1)$. Then $M^\lambda \cong \mathbb{C}^n$. Then $\text{Ind}_{S_\lambda}^{S_n} \text{Triv}$ is the character of the permutation representation.

Definition 3.1.3 (Young Diagram)
Let $n \in \mathbb{N}^+$. Let $\lambda = (\lambda_1, \dots, \lambda_k)$ be a partition of n. Define the Young diagram of λ to be

$$Y(\lambda) = \{(i,j) \in \mathbb{N}^+ \times \mathbb{N}^+ \mid 1 \leq i \leq k, 1 \leq j \leq \lambda_i\}$$

Example 3.1.4
Consider the partition $(4, 3, 1)$ of 8. The Young diagram of the partition is given by



Definition 3.1.5 (Young Tableau)
Let $n \in \mathbb{N}^+$. Let λ be a partition of n. A Young tableau of λ is a bijection $t : Y(\lambda) \to \{1, \dots, n\}$. Denote the set of all Young tableau of λ by Δ^λ.

Definition 3.1.6 (Young Tabloid)
Let $n \in \mathbb{N}^+$. Let λ be a partition of n.
- Let t_1, t_2 be Young tableau of λ. We say that t_1 and t_2 are equivalent if
$$\{t_1(r_0, j) \mid 1 \leq j \leq \lambda_i\} = \{t_2(r_0, j) \mid 1 \leq j \leq \lambda_k\}$$
for all $1 \leq r_0 \leq k$ (in other words the rows of the tableau are equal up to permutation).
- A Young tabloid of λ is an equivalence class $[t]$ where t is a Young tableau of λ.

Example 3.1.7
The following Young tableaux of $(3, 2)$ are equivalent.
$$\begin{array}{|c|c|c|}\hline 2 & 1 & 4 \\ \hline 5 & 3 & \\ \cline{1-2}\end{array} \sim \begin{array}{|c|c|c|}\hline 1 & 4 & 2 \\ \hline 3 & 5 & \\ \cline{1-2}\end{array} \sim \begin{array}{|c|c|c|}\hline 1 & 2 & 4 \\ \hline 3 & 5 & \\ \cline{1-2}\end{array}$$

Definition 3.1.8 (The Associated Polytabloid of a Tableau)
Let $n \in \mathbb{N}^+$. Let λ be a partition of n. Let t be a λ-tableau. Define the associated polytabloid to be
$$e(t) = \sum_{\sigma \in C(t)} \text{sign}(\sigma)(\sigma \cdot [t])$$
where $C(t) = \prod_{i=1}^k \text{Sym}(C_i)$ where C_i is the ith column of t.

Example 3.1.9
Consider the following Young tableau t:
$$\begin{array}{|c|c|c|}\hline 2 & 3 & 5 \\ \hline 1 & 4 & \\ \cline{1-2}\end{array}$$
Then $e(t)$ is the sum of four terms.

Definition 3.1.10 (The Specht Module of a Partition)
Let $n \in \mathbb{N}^+$. Let λ be a partition for n. Define the Specht module associated to λ to be the $\mathbb{C}$-vector space
$$S^\lambda = \mathbb{C}\langle e(t) \mid t \text{ is a } \lambda\text{-tableau } \rangle$$

Proposition 3.1.11
Let $n \in \mathbb{N}^+$. Then $\{S^\lambda \mid \lambda \text{ is a partition for } n\}$ forms the complete list of non-isomorphic irreducible $\mathbb{C}[S_n]$-modules.